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A Machine Learning Approach for Effective Malaria Diagnosis and Parasite Profiling

Aiden Yang
Seven Hills School
aidenyang2010@gmail.com

Kate Plas
Department of Computer Science
University of Cincinnati
plaska@mail.uc.edu

Justin Zhan
Department of Computer Science
University of Cincinnati
zhanjt@ucmail.uc.edu

Abstract—Malaria is a mosquito-borne disease transmitted by the bite of female *Anopheles* mosquitoes infected with parasites of the *Plasmodium* genus. Each year, malaria causes approximately 600,000 deaths worldwide and accounts for over 260 million clinical cases. Because the disease disproportionately affects remote regions with limited access to medical infrastructure, there is a critical need for affordable, accessible, and non-invasive methods that enable early detection and effective treatment. To address this need for scalable diagnostic tools, we analyze state-of-the-art machine learning approaches for malaria identification and characterization from microscopic blood smear images. Our results show that YOLOv12 consistently outperforms other advanced architectures, making it an optimal choice for both differentiating healthy from parasitized red blood cells and for characterizing parasite species and morphology through segmentation. Furthermore, this work establishes the first known segmentation benchmarking results on the MP-IDB dataset using recent YOLO models and provides updated baselines for malaria diagnostic object detection methods.

Index Terms—object detection, classification, segmentation, YOLO, microscopy

I. INTRODUCTION

Malaria is a mosquito-borne disease transmitted by the bite of female *Anopheles* mosquitoes infected with parasites of the genus *Plasmodium*. Of the 156 named *Plasmodium* species known to infect vertebrates, four are responsible for human malaria: *P. falciparum*, *P. vivax*, *P. ovale*, and *P. malariae* [1]. Among these, *P. falciparum* is the most prevalent and accounts for the majority of malaria-related deaths.

The World Health Organization (WHO) publishes an annual malaria report to assess global disease burden and progress toward control and eradication. In 2023, the WHO estimated 263 million malaria cases and 597,000 deaths across 83 countries, placing nearly half of the world's population at risk [2]. The impact of malaria is highly uneven: in 2022, approximately 95% of deaths occurred in the WHO African region, with nearly 80% affecting children under five years of age living in sub-Saharan Africa.

Clinically, malaria presents with non-specific, flu-like symptoms such as fever, headache, chills, fatigue, nausea, and diarrhea. Severe cases may progress to impaired consciousness, convulsions, respiratory distress, and abnormal bleeding. The incubation period typically ranges from 7 to 30 days; however, parasites, particularly *P. vivax*, can remain dormant in the liver

for months or even years before re-entering the bloodstream and causing relapse [3].

Because malaria disproportionately affects remote regions with limited access to medical infrastructure, early detection through affordable, accessible, and non-invasive diagnostic methods is critical for effective treatment. Symptom overlap with other febrile illnesses and the parasite's capacity for dormancy significantly reduce diagnostic accuracy in the absence of laboratory confirmation.

The current gold standard for malaria diagnosis and characterization involves microscopic examination of stained blood smears to identify infected cells. While reliable, this approach requires trained personnel and specialized equipment. Polymerase chain reaction (PCR)-based presence-absence tests offer comparable accuracy but incur higher costs and demand technical expertise, limiting their practicality in low-resource settings. These personnel and technological constraints have motivated growing interest in automated object detection and classification methods as scalable alternatives for improving malaria diagnostic accessibility [4].

To address these challenges, automated diagnosis via blood smear image analysis offers a promising approach. Object detection is the process of classifying and localizing entities of interest by drawing bounding boxes or segmentation masks around them. For malaria diagnosis, infected cells are differentiated from non-parasitic stained components/bodies (e.g., RBCs, white blood cells, platelets, and artifacts). A positive diagnosis is given if there are parasites present in the blood slide. To further characterize the infection, a detector must also be capable of differentiating parasite species and life stages [5]. Figure 1 shows, on the left, bounding boxes located on infected cells for malaria diagnosis, and on the right, individual parasite segmentation masks and species classes for infection characterization. Figure 2 shows the wide variety of blood slide preparation methods (smear type and stain type) as well as the heterogeneity of parasite stage and type.

Object detection architectures can be broadly categorized into two-stage and single-stage models. Single-stage detectors directly regress bounding boxes and class probabilities in a single feed-forward pass, while two-stage detectors first generate region proposals and subsequently classify and refine them. Two-stage approaches have historically achieved higher accuracy due to their explicit proposal refinement, but recent

advances in single-stage detectors, particularly within the You Only Look Once (YOLO) family, have substantially narrowed this performance gap [6].

With these improvements, single-stage detectors have become increasingly attractive for malaria diagnosis, where rapid localization and classification are essential, and computational resources are often limited. Their low-latency inference and reduced architectural complexity make them well-suited for deployment in resource-constrained clinical environments. Among single-stage approaches, YOLO models have emerged as leading solutions due to their favorable trade-off between accuracy and efficiency [7].

The YOLO family (v1–v11) comprises real-time, single-stage object detection models originally built on convolutional neural networks [8]. The name “You Only Look Once” reflects the model’s unified design, in which object localization and classification are performed jointly in a single network evaluation over the entire image. This condensed pipeline enables significantly faster inference than two-stage detectors such as R-CNN, which rely on computationally expensive region proposal mechanisms. Prior studies have explored YOLO-based approaches for malaria detection, including the use of YOLOv2 combined with an AlexNet classifier for identifying *P. vivax* [9], as well as evaluations of YOLOv3 and YOLOv4 on binary malaria classification tasks [4]. Abdurahman et al. [4] investigated YOLOv4 because it operates efficiently on conventional GPUs, but also showed the modified version’s superior accuracy [10]. More recent YOLO variants introduce capabilities that are particularly relevant for microscopy imaging. YOLOv8 was the first version to support instance segmentation in addition to classification and bounding box prediction [11]. YOLOv12 further advances the architecture by transitioning from a purely convolutional backbone to an attention-centric design while maintaining real-time performance [7]. Its area attention (A2) module enables selective focus on salient image regions, improving feature extraction and localization of small objects—an important requirement for detecting malaria parasites in blood smears. Despite these advances, modern YOLO architectures have not yet been systematically evaluated for multi-species malaria segmentation and classification.

A. Related works

Several machine learning (ML) methods have been applied for malaria-infected RBC classification tasks. Traditional methods using support vector machine (SVM) and linear discriminant analysis (LDA) suffer from data availability and quality constraints [12]. Deep learning approaches using CNNs have been implemented to learn feature representations from microscopy images. Liang et al. reported 97.37% accuracy using their 16-layered convolutional neural network (CNN) model to perform classification on archived blood smear images acquired from Chittagong Medical College Hospital, Bangladesh [13]. This model focuses on the classification task of already segmented red blood cells (RBCs) as either infected or not, using a CNN. Because microscopy images generally

contain large numbers and diverse types of cells, this limits full applicability without integration of object detection model capabilities for further detailed analysis on parasite species, life stage, proportions of infected RBCs, and morphological data.

Attention-Dense Circular Net (ADCN) [14], deep belief-based classification networks [15], LeNet [16], AlexNet [17], and GoogleNet [18], have been developed to identify RBCs infected with malaria parasites using the combined Malaria Cell Images Dataset from the NIH, made from five datasets [19]–[23]. The Dilated Attention Network (DANet) was recently proposed as a lightweight architecture incorporating dilated attention to better capture contextual information in low-contrast blood smear images [24]. On the NIH Malaria Cell Images Dataset, DANet achieves 97.95% classification accuracy and a 97.86% F1-score while using only 2.3 million parameters. However, its evaluation is limited to a binary, single-species dataset, reflecting a broader limitation of existing approaches: most methods focus solely on infected versus uninfected cell classification for a single parasite species, despite substantial variation in parasite morphology across species and life stages [5].

To address dataset class-label limitations, the MP-IDB dataset was introduced with multiclass labels for parasite species and developmental stages [5]. Nevertheless, this dataset remains underexplored. While approaches for multiclass classification, such as Hybrid CapNet, report strong classification performance, they do not employ the instance-level segmentation masks provided in MP-IDB, thereby omitting valuable information for analyzing parasite morphology and infection severity [25].

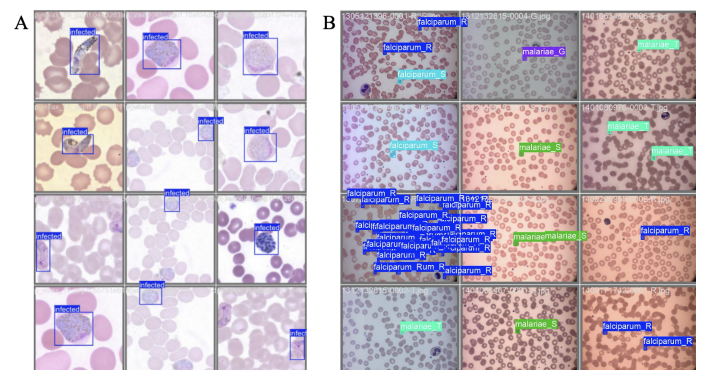


Fig. 1. (A) Set of binary classification labels and bounding boxes on blood slide images from the Malaria Detection Computer Vision Dataset [26] (B) Sample of multiclass classification labels, bounding boxes, and segmentation masks from MP-IDB [5]. Note that MP-IDB annotations are by-parasites rather than by-cell.

Contributions: We demonstrate that YOLOv12 achieves strong performance across both binary and multiclass malaria diagnostic tasks, including infected cell identification, parasite detection, species classification, and instance segmentation. To our knowledge, this work presents the first systematic comparison of YOLO architectures for segmentation using the MP-IDB multiclass dataset. Additionally, we provide an

updated performance analysis of contemporary YOLO variants, establishing a current benchmark for automated malaria classification and parasite characterization. Specifically, (1) we evaluate YOLOv12 on binary and multiclass malaria diagnostic tasks, demonstrating strong performance across detection, classification, and segmentation; (2) we present the first systematic benchmarking of YOLO-based segmentation models on the MP-IDB multiclass dataset; and (3) we establish updated baselines for automated malaria classification and parasite characterization using modern YOLO variants.

II. METHODS

We evaluate two diagnostic tasks. First, we perform binary classification to determine whether a given cell is infected. Second, we evaluate the segmentation and multiclass classification capabilities of recent YOLO architectures for differentiating parasite species and developmental life stages. All YOLO models were implemented using the Ultralytics Python library [7], [27]. Experiments were conducted on a system equipped with four NVIDIA A100 SXM4 GPUs.

A. Binary classification

Models were trained to perform binary classification on two datasets (Malaria Detection Computer Vision Dataset and Compiled Malaria Smartphone Data) totaling 363 blood slide images of different scales [26], [28]. The annotations contained bounding box coordinates and labels for the infected cells. Images were partitioned into train, validation, and test subsets, accounting for 80%, 10%, and 10% of the total, respectively. To ensure consistent performance across images captured at different magnification levels, we applied scale augmentation during training. Hyperparameters (shown in Table I) were tuned over 40 epochs with 100 iterations to determine the optimal settings for each model.

Hyperparameter	YOLOv8	YOLOv11	YOLOv12
Model and Training Hyperparameters			
Initial Learning Rate (lr0)	0.01214	0.00799	0.00768
Final Learning Rate Factor (lrf)	0.01397	0.01019	0.0129
Momentum	0.88008	0.8681	0.92159
Weight Decay	0.00046	0.00059	0.00051
Warmup Epochs	3.52352	2.83424	3.12773
Warmup Momentum	0.53738	0.61555	0.75943
Box Loss Gain	6.15276	6.50232	6.9982
Class Loss Gain	0.76696	0.44618	0.49876
DFL Loss Gain	1.33094	1.68421	1.65784
Data Augmentation Hyperparameters			
HSV Hue Augmentation	0.01126	0.01468	0.00805
HSV Saturation Augmentation	0.82266	0.9	0.86878
HSV Value Augmentation	0.47008	0.36375	0.51735
Translation	0.0452	0.15679	0.14421
Scaling	0.25091	0.35667	0.46391
Horizontal Flip	0.5621	0.3859	0.50508
Mosaic Augmentation	0.83081	0.85663	0.68666

TABLE I

OPTIMAL HYPERPARAMETERS FOUND DURING TUNING FOR YOLOv8, YOLOv11, AND YOLOv12.

B. Multiclass classification

MP-IDB's 209 full-slide blood images contain four malaria parasite types: *P. ovale* (N=29), *P. malariae* (N=37), *P. vivax* (N=40), and *P. falciparum* (N=103), in addition to the distinct stage (schizont development, ring phases, trophozoite growth, and gametocyte growth). We specifically acquire images and annotations from the MP-IDB-YOLO dataset, which is a version of MP-IDB already in YOLO format [29].

Class	Images	Objects	Avg count	Avg area
<i>P. falciparum</i>	103	1267	12.3	0.79%
<i>P. vivax</i>	40	64	1.6	0.35%
<i>P. malariae</i>	37	43	1.16	0.14%
<i>P. ovale</i>	29	33	1.14	0.22%

TABLE II

MP-IDB STATISTICS FOR MALARIA SPECIES DETECTION AND SEGMENTATION [5].

Because of the wide range of parasite species, developmental stage, imaging technology, and blood slide preparation method, a generalized malaria-parasite detection model remains difficult to design [30]. To address the heterogeneity as well as the class imbalance favoring *P. falciparum*, we augment the MP-IDB dataset using a number of photometric and geometric transformations. Hue, saturation, and value jittering introduce image perturbations in stain color, contrast, and brightness, to reflect the differences in slide preparation, lighting, and cameras. Small random scaling is applied to simulate magnification inconsistencies. Vertical and horizontal flips add further image variety.

We analyze and perform multiclass classification on four classes for each of the parasite species. Images were partitioned into train, validation, and test subsets (80-10-10).



Fig. 2. Range of blood slide dataset variability due to differences in smear type, stain type, parasite species, and life stage (adapted from [30]).

C. Evaluation metrics

We evaluate localization performance using mean Average Precision (mAP) for both bounding boxes and segmentation masks. Object detection and instance segmentation rely on the Intersection over Union (IoU) metric, which quantifies the overlap between a predicted region P and the ground-truth region G :

$$\text{IoU}(P, G) = \frac{|P \cap G|}{|P \cup G|}.$$

A prediction is considered correct if its IoU with the ground truth exceeds a specified threshold (e.g., 0.50 or 0.75). Average Precision (AP) summarizes the area under the precision-recall curve for an IoU threshold. For a dataset with N images, AP for a region type (bounding box or segmentation mask) is defined as

$$\text{AP}_{\text{area}} = \frac{1}{N} \sum_{i=1}^N \text{AP}_{\text{area}}^{(i)},$$

where $\text{AP}_{\text{area}}^{(i)}$ is the average precision computed on the i -th image. Mean Average Precision (mAP) averages AP across the IoU thresholds (e.g., mAP@0.50 or mAP@0.50-0.95).

Classification performance was measured using standard metrics, including precision, recall, F1 score, and training loss. These metrics are used to evaluate the model's ability to distinguish infected from uninfected cells correctly, as well as to differentiate among parasite species and life stages in the multiclass setting.

III. RESULTS

A. Binary classification with bounding box localization

On the 177th epoch of training, YOLOv12 achieved a precision of 98.137%, a recall of 92.025%, an mAP50 of 0.94566, and an mAP50-95 of 0.60753 (shown in Table III). It scored a box loss of 1.3714, a cross-entropy loss of 0.52217, and a distribution focal loss of 0.94913. Performing on the validation set, the model scored a box loss of 1.1930, a cross-entropy loss of 0.81145, and a distribution focal loss of 1.1109. These metrics stayed relatively constant for the rest of the training. The model maintains a high F1 score of around 0.95 with a confidence threshold less than or equal to 0.6.

Model	Precision	Recall	F1 Score	Loss
YOLOv12	0.9814	0.9202	0.9498	0.5222
DenseNet121	0.9314	0.9652	0.9457	0.1487
DenseNet169	0.9327	0.9659	0.9485	0.1602
CNN	0.9377	0.9826	0.9591	0.2754
InceptionV3	0.8206	0.9129	0.8592	0.3737
VGG16	0.9253	0.9804	0.9518	0.4253
ResNet50	0.8751	0.9761	0.9195	0.2911
EfficientNet-B7	0.9424	0.9862	0.9639	0.1105
MobileNet	0.9302	0.9775	0.9522	0.2212
MobileNetV2	0.8663	0.9028	0.8811	0.2842

TABLE III

MODEL PERFORMANCE METRICS FOR BINARY CLASSIFICATION ON THE COMBINED MALARIA DETECTION COMPUTER VISION AND COMPILED MALARIA SMARTPHONE DATASETS [26], [28].

B. Multi-label classification with segmentation masks

1) *Segmentation evaluation*: Among the three models, YOLOv12 achieved the highest sustained mAP@0.50 throughout training, ending with mAP@0.50 = 0.82. YOLOv11 followed closely, whereas YOLOv8 showed the lowest plateau (at mAP@0.50 = 0.77) and the greatest variance. Under the stricter mAP@0.50-0.95 metric, for fine-grained segmentation accuracy evaluation at higher IoU thresholds, the performance gap between models became more pronounced. YOLOv12 achieved the most stable long-term performance, while YOLOv11 produced early peaks but failed to sustain high accuracy across epochs. YOLOv8 gain lagged behind both, with lower average performance and higher noise across training.

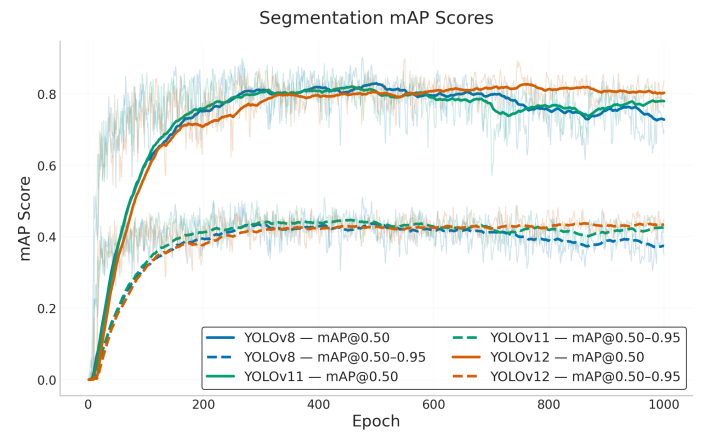


Fig. 3. Mean average precision (mAP) between YOLO models for mAP@0.50 and mAP@0.50=0.95 over 1000 training epochs.

2) *Species classification evaluation*: Species-level classification performances vary between YOLO models. For *P. falciparum*, YOLOv8 achieves 92% correct classification with 17% misclassified as background, while YOLOv11 maintains 92% correct classification with 12% predicted as background; YOLOv12 attains 95% correct classification with 5% predicted as background. *P. Vivax* is classified with 100% accuracy by YOLOv8 and YOLOv11, while YOLOv12 shows reduced performance with 75% correct predictions and 25% predicted as background. *P. Ovale* achieves 100% correct classification in YOLOv8 and YOLOv11, with YOLOv12 recording 100% accuracy and 3% misclassification into background. For *P. malariae*, YOLOv8 attains 78% correct predictions with 22% labeled as background, YOLOv11 matches this 78% accuracy with 22% misclassified, and YOLOv12 reaches 89% correct classification with 11% misclassified as background. The background category includes non-parasitic regions, including uninfected RBCs, leukocytes, artifacts, and unstained slide areas. YOLOv8 assigns 8% of background instances incorrectly to parasite classes, YOLOv11 assigns 8%, and YOLOv12 assigns 11%.

IV. DISCUSSION

For binary classification, YOLOv12 outperforms many existing models on blood smear images, achieving a precision of 98.137% and a recall of 92.025%. In comparison, InceptionV3 reports a precision of 82.06% and a recall of 91.2%, while AIDMAN achieves a precision of 97.13% and a recall of 97% on thin blood smear images. Although YOLOv12 demonstrates strong precision, the observed cross-entropy loss of 0.52 during training suggests room for further improvement, which may be addressed through dataset expansion and improved class balance in future experiments. It is important to note that these metrics for binary classification are reported on different datasets and experimental settings, and direct comparisons should therefore be interpreted with caution. Rigorous clinical validation would be required to assess relative performance across models more accurately. In future work, we plan to continue fine-tuning the model to improve sensitivity while further reducing training loss.

In segmentation tasks, YOLOv12 consistently outperforms earlier YOLO variants. This improvement is likely attributable to architectural refinements that enhance robustness to scale variation and improve small-object localization. Specifically, YOLOv12 incorporates strengthened multi-scale feature fusion, a redesigned mask head with improved boundary sensitivity, and refined loss functions that better optimize mask accuracy. Collectively, these enhancements lead to higher-quality segmentation masks, improved mAP₅₀₋₉₅, and stronger performance on small, low-contrast objects. Relative to models like DANet and Hybrid CapNet, YOLOv12 demonstrates superior binary classification of malaria-infected RBCs. Our trained YOLOv12 model adds another layer of analysis by being able to further identify a variety of parasites.

For multiclass classification, YOLOv12 demonstrates the most balanced overall performance across parasite species. It achieves the highest classification accuracy for *P. falciparum* (85.9%) and perfectly discriminates both *P. vivax* and *P. ovale*. In addition, YOLOv12 produces the fewest background false positives, reducing non-parasitic misclassifications by more than 50% compared to YOLOv11. YOLOv11 exhibits mixed performance across species, with notable cross-species confusion, particularly misclassifying *P. ovale* as *P. vivax*, and generates the highest number of false parasite detections in background regions. YOLOv8 shows generally stable performance without cross-species confusion but suffers from elevated false negatives for *P. falciparum* and *P. malariae*, indicating limitations in distinguishing low-contrast parasitic forms. Across all evaluated models, *P. malariae* remains the most challenging species, frequently misclassified as background.

V. CONCLUSION

This benchmarking work demonstrates that recent YOLO architectures, particularly YOLOv12, can substantially improve malaria diagnosis and parasite segmentation across heterogeneous blood smear datasets. Among all evaluated models, YOLOv12 achieved the strongest overall performance in terms

of accuracy, stability, and localization, outperforming classical convolutional neural networks and earlier YOLO variants on both segmentation and classification tasks. These findings highlight the potential of real-time, attention-enhanced detectors to enable scalable and accessible malaria diagnostics in resource-constrained clinical environments.

Despite these advances, challenges remain due to blood smear heterogeneity, limited annotation quality, and pronounced class imbalance. As future work, we plan to explore augmenting training data with synthetic samples generated by generative image models to improve robustness and generalization. Additionally, we aim to investigate the inclusion of other cell types to better characterize the immune microenvironment surrounding malaria parasite-infected cells.

Data Availability and Reproducibility: Code and the models' trained weights are available and can be downloaded at <https://github.com/plask3189/malariaWeights.git>. All image datasets used for training and evaluation are publicly accessible.

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